

National Exam of Higher National Diploma 2026 Session

Specialty/option: SOFTWARE ENGINEERING (SWE)
Paper : Computer Technology
Duration : 4 hours

Credits : 6

GENERAL INSTRUCTIONS.

- **ANSWER ALL QUESTIONS**
- *You are reminded of the necessity of good English and orderly presentation of your material.*

SECTION A: MULTIPLE CHOICE QUESTIONS (50 MARKS)

One Mark (01) For Each Correct Answer

1. Type of data arrangement most commonly found in traditional databases is:
 - A. Big data
 - B. Structured data
 - C. Unstructured data
 - D. Data warehouse
2. The purpose of normalization is _____.
 - A. To eliminate redundant data
 - B. To provide a reason to denormalize a database
 - C. To increase the numbers of relations
 - D. To optimize data for insert, update, and delete.
3. The three kinds of anomalies are _____.
 - A. Insertion, Selection, Deletion
 - B. Insertion, Update, Deletion
 - C. Selection, Update, Deletion
 - D. Creation, Read, Update
4. Which of the following is not a valid SQL aggregate function?
 - A. SUM
 - B. AVG
 - C. SUB
 - D. MAX
5. The insertion anomaly refers to a situation in which _____.
 - A. Data must be inserted before it can be deleted.
 - B. Too many inserts cause the table to fill up.
 - C. Data must be deleted before it can be inserted.
 - D. A required insert cannot be done due to an artificial dependency.

6. Which of the following statements is NOT valid about the ALTER clause?
- A. May be used to add a constraint
 - B. May be used to drop a constraint
 - C. May be used to add a view
 - D. May be used to drop a table column
7. An UPDATE statement without a WHERE clause will _____
- A. result in an error message
 - B. Update no row in a table
 - C. Update every row in a table
 - D. update all the tables in the Database
8. In an instance of a relation, the number of tuples is known as _____
- A. Degree
 - B. Aggregate
 - C. Cardinality
 - D. BTotal rows
9. The machine that runs the Database Management System is referred to as _____
- A. Database server
 - B. Business Logic
 - C. Client
 - D. None of the above.
10. Users interaction with the database must be routed through _____
- A. File Management System
 - B. Database Management System
 - C. Client/Server System
 - D. Transaction processing System
11. Which device would primarily function at the Data Link layer?
- A. Routers
 - B. Firewalls
 - C. Gateways
 - D. Switches
12. Which is the proper order of the OSI layers?
- A. Application, Transport, Session, Presentation, Network, Data Link, Physical
 - B. Presentation, Application, Session, Transport, Network, Data Link, Physical
 - C. Application, Presentation, Session, Transport, Network, Data Link, Physical
 - D. Application, Presentation, Transport, Network, Session, Data Link, Physical
13. Which OSI layer is responsible for logical addressing?
- A. Transport layer
 - B. Network layer
 - C. Application layer
 - D. Data Link layer
14. Which network topology design has a centralized switch connecting all of the devices?
- A. Star topology
 - B. Full mesh topology

- C. Partial mesh topology
 - D. Hybrid topology
15. Which cable type would you use to connect a router to a switch?
- A. Straight-through cable
 - B. Crossover cable
 - C. Rolled cable
 - D. Shielded cable
16. Which method is used to direct communications to a group of computers that subscribe to the transmission?
- A. Unicast
 - B. Broadcast
 - C. Multicast
 - D. Anycast
17. Which protocol is used to assign automatic IP address to all hosts in a network?
- A. A) DHCP
 - B. DNS
 - C. ARP
 - D. FTP
18. Which protocol on the host machine allows for a mapping of IP to MAC address?
- A. MAC Address Resolution Protocol
 - B. Address Resolution Protocol
 - C. Reverse Address Resolution Protocol
 - D. Internet Control Message Protocol
19. Which criteria are routing decisions based upon?
- A. Source IP
 - B. Destination IP address
 - C. TTL
 - D. Destination MAC address
20. Which is a direct benefit of a full mesh topology?
- A. Increased bandwidth
 - B. Increased redundancy
 - C. Decreased switch count
 - D. Increased complexity
21. running time complexity of a linear algorithm is given as _____
- A. $O(1)$
 - B. $O(n)$
 - C. $O(n \log n)$
 - D. $O(n^2)$
22. Which among the following is a LIFO data structure?
- A. Stack
 - B. Linked list
 - C. Queue
 - D. Graph
23. Which of the following cases occurs when searching an array using linear search the

value to be searched is equal to the first element of the array?

- A. Worst case
- B. Average case
- C. Best case
- D. Amortized case

24. In a _____ sort, consecutive adjacent pairs of elements in the array are compared with each other.

- A. Bubble
- B. Selection
- C. Merge
- D. Radix

25. Which of the following techniques deals with sorting the data stored in the computer's memory?

- A. Insertion sort
- B. Internal sort
- C. Merge sort
- D. External sort

26. _____ design approach starts by dividing the complex algorithm into one or more modules.

- A. Bottom up
- B. Modular
- C. Top down
- D. Smart

27. A complex algorithm is often divided into smaller units called _____.

- A. Blocks
- B. Functions
- C. links
- D. Modules

28. A(n) _____ node has zero degree.

- A. leaf
- B. internal
- C. isolated
- D. Graph

29. The memory use of an adjacency matrix is _____.

- A. $O(n)$
- B. $O(n^2)$
- C. $O(n^3)$
- D. $O(\log n)$

30. An edge that has identical end-points is called a _____.

- A. Multi-path
- B. Loop
- C. Cycle
- D. Multi-edge

31. Consider the array declaration

```
int Num[] = {0, 2, 4, 6, 8, 10, 12, 14, 16};
```

The value of Num[Num[3]] is _____

- A. 4
- B. 6
- C. 12
- D. 16

32. What is the output of the code snippet below?

```
int main()
{
    float c= 3.14;
    printf("%f", c%2);
    return 0;
}
```

- A. 157
- B. 1.57
- C. 3
- D. Compiler error

33. What is the output of the code snippet below?

```
int main(void)
{
    int x=1,y=5;
    printf("%d ", ++x+y);
    return 0;
}
```

- A. 6
- B. 7
- C. 8
- D. Compiler error

34. By default, when a C program calls the function, argument is passed by _____

- A. reference
- B. parameter
- C. value
- D. type

35. Which of the following declarations is NOT valid in C?

- A. float NumberOfStudents;
- B. char LetterOfAlphabet;
- C. double CelsiusTemperature;
- D. int 5Students;

36. If X is an integer variable, $X = 5/2$ will return the value:

- A. 2.5
- B. 3

- C. 2
D. 0
37. Operator precedence determines which operator _____
A. is the most important
B. is used first
C. is the fastest
D. operates on largest scale
38. Flow charts are used to _____
A. Decide sequence of steps involved in finding a solution
B. Aid in making algorithm
C. Prepare decision tables
D. Debug a program
39. Which statement must NOT end with a semicolon?
A. #define
B. variable declaration
C. assignment
D. none
40. A location in the computer's memory that may contain different values at various times throughout the execution of a program is called a _____
A. Constant
B. Variable
C. Cache
D. Register

SECTION B (60 MARKS)

Part I: Programming (20 Marks)

- a) Find the error in each of the following C code segments and explain how to correct it.
- a) `x = 1;`
 `while ( x <= 10 );`
 `++x;`
 `}`
b) `for ( y = .1; y != 1.0; y += .1 )`
 `printf( "%f\n", y );`
- (2 marks)
(2 marks)
- b) The code segment below represents a function written in C.

```

int sum( int n )
{
    if ( x == n ) {
        return x; //
    }
    else {
        n + sum( n - 1 );
    }
}

```

- a) Name the kind of function represented by the code fragment above. (1 mark)
- b) The above function will return an improper result. What must be done for it to return the right result? (1 mark)
- c) Consider a 2-by-5 integer array t.
 - i) Declare the array t. (2 marks)
 - ii) How many rows does t have? (1 mark)
 - iii) How many columns does t have? (1 mark)
 - iv) How many elements does t have? (1 mark)
 - v) Write the names of all the elements in the second row of t. (2 marks)
 - vi) Write a single statement that sets the element of t in row 1 and column 2 to zero. (2 marks)
- d) i) What would you use in a C program to store data permanently in a computer? (1 mark)
- ii) What is a stream in C? (1 mark)
- iii) What are the four actions to be implemented when using a file? (2 marks)
- iv) What will happen if you don't close a file? (1 mark)

Part II: Data Structure and Algorithm (10 Marks)

- 1) State the two basic properties of greedy algorithms. (2 marks)
- 2) Given the following adjacency matrix, draw the weighted graph.

$$\begin{pmatrix}
 0 & 4 & 0 & 2 & 0 \\
 0 & 0 & 0 & 7 & 0 \\
 0 & 5 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 3 \\
 0 & 0 & 1 & 0 & 0
 \end{pmatrix}$$

(4 marks)
- 3) The Dijkstra's algorithm is such that, given a graph G and a source node Q, the algorithm is used to find the shortest path (one having the lowest cost) between Q (source node) and every other node reachable from Q.
 Consider the graph, G, as shown in Figure 2. Taking Q as the initial node, execute the Dijkstra's algorithm on it.

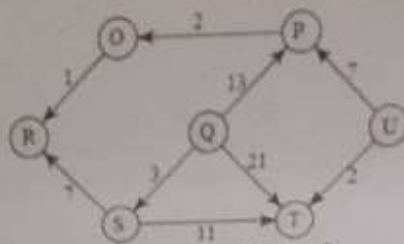


Figure 2

(4 marks)

Part III: Database (15 Marks)

1) Define the following terms:

- Metadata
- Database
- Database Management System
- Entity
- Attribute

(5 marks)

2) Consider the Entity-Relationship diagram in Figure 4. Describe the business rule that exist

between the following entities (for example; consider SUPPLIER to ITEM, and ITEM to SUPPLIER):

- SUPPLIER and ITEM
- ITEM and PRODUCT
- CUSTOMER and ORDER.

(2 x 3 marks)

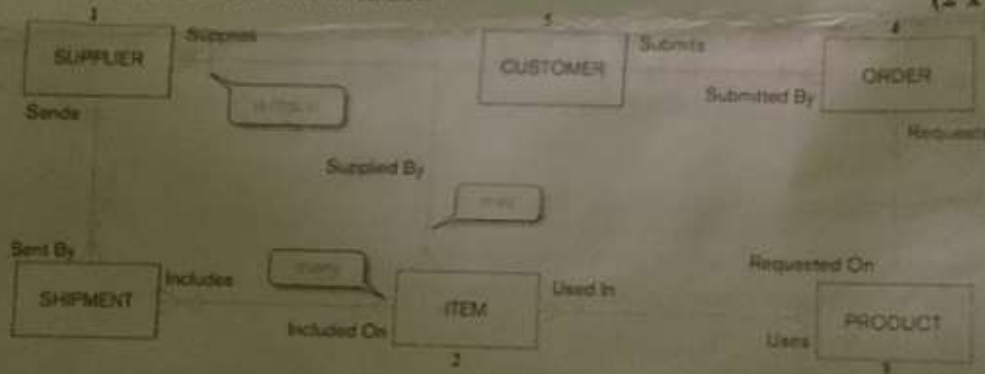


Figure 4

3) State four properties of Relations.

(4 marks)

Part IV: Networking (10 Marks)

1) Give three (03) network control devices?

(1.1marks)

2) Differentiate but a client server and a peer-to-peer network architecture.

(4 marks)

3) Identify the following Protocols, give their functions and the layer they are found on the TCP/IP model.

- ARP
- UDP
- HTTP

(4.5 marks)